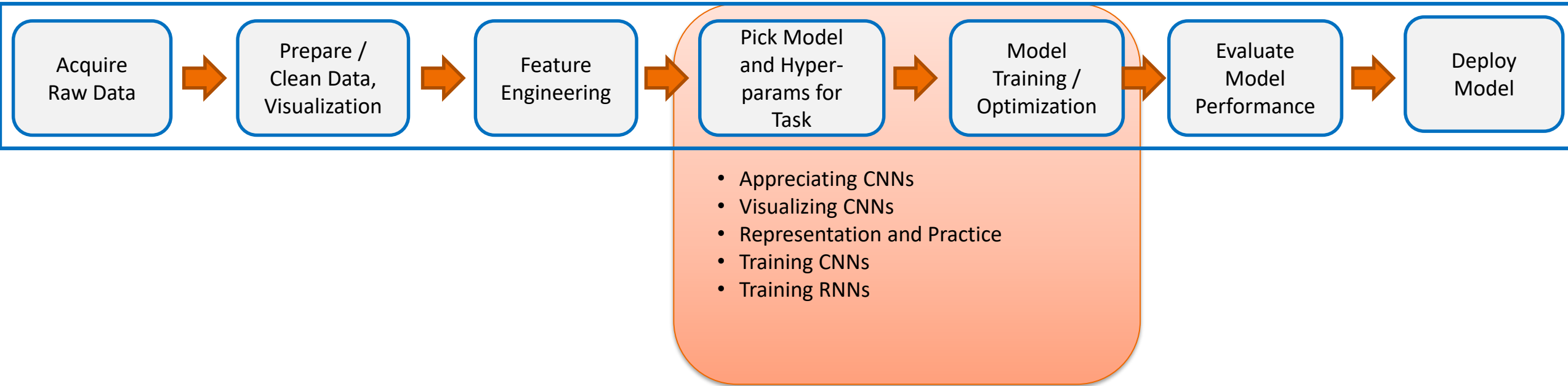


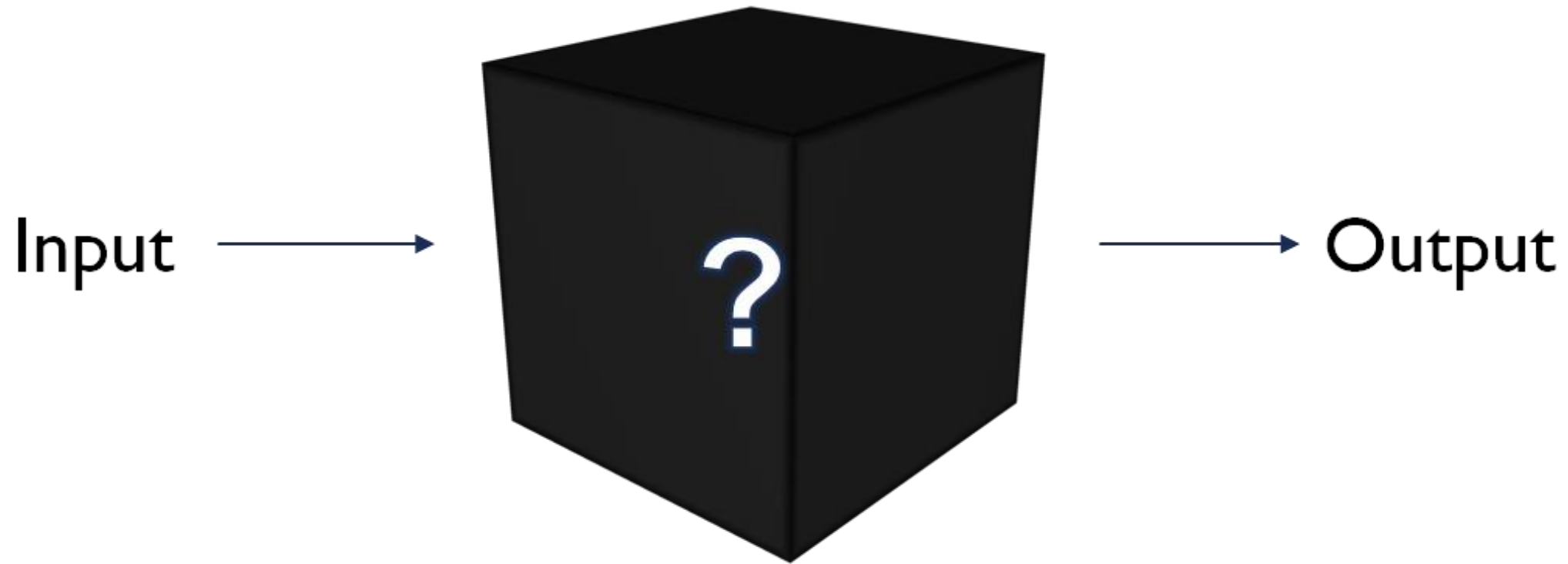
Today's Focus



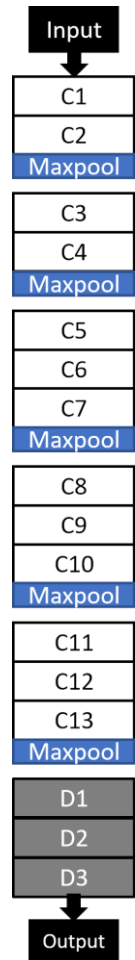
Explainability and Interpretability

— Visualization, Interpretation —

What goes on inside a convnet?



CNN Visualizations



What is in this image?



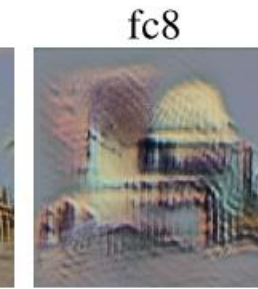
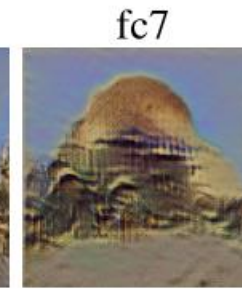
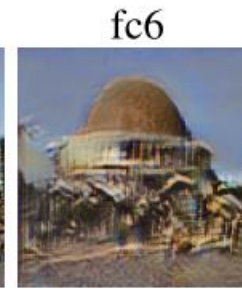
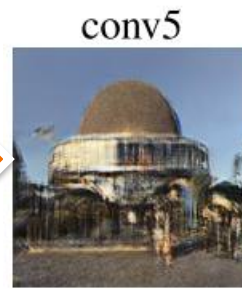
Saliency

Cat

Dog

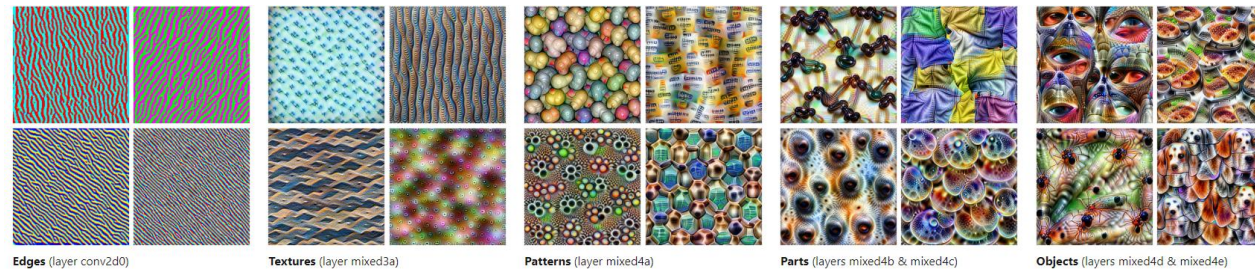
Which part of the image explains the classification?

Feature Inversion



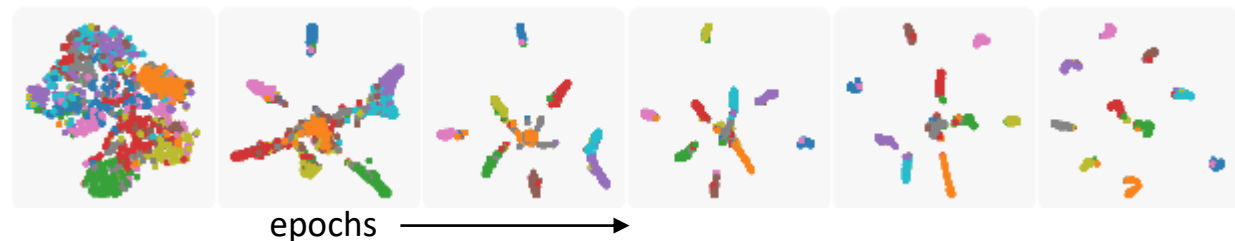
How much information is retained/discarded at each layer of a deep network?

Feature visualization



What pattern does each convolutional filter search for?

Monitoring training



How well is the CNN getting trained?

- Activation maximization
- Deep Dream
- Feature inversion
- Saliency visualization
- Statistical methods

How do we study what happens inside convnets

LOOK INSIDE CONVNETS

ACTIVATION MAXIMIZATION

Activation Maximization

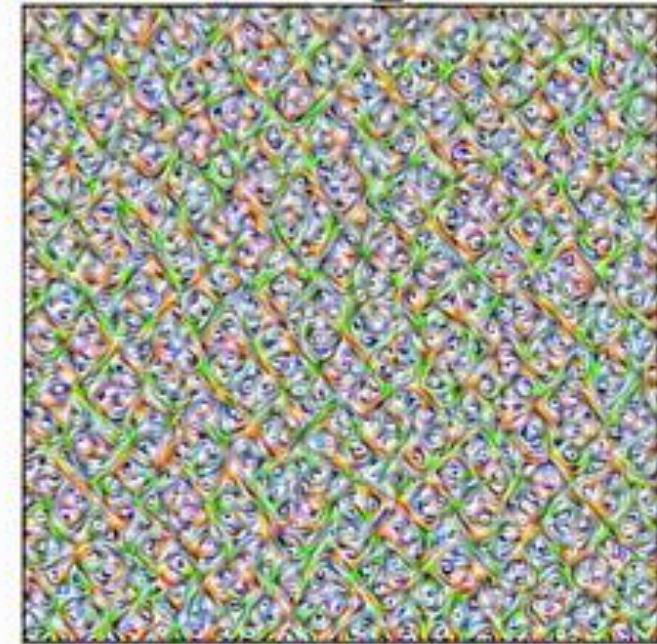
- Visualize filters of a conv net
- What patterns is a filter looking for in an input image?

Filters get more complicated
towards the last layers

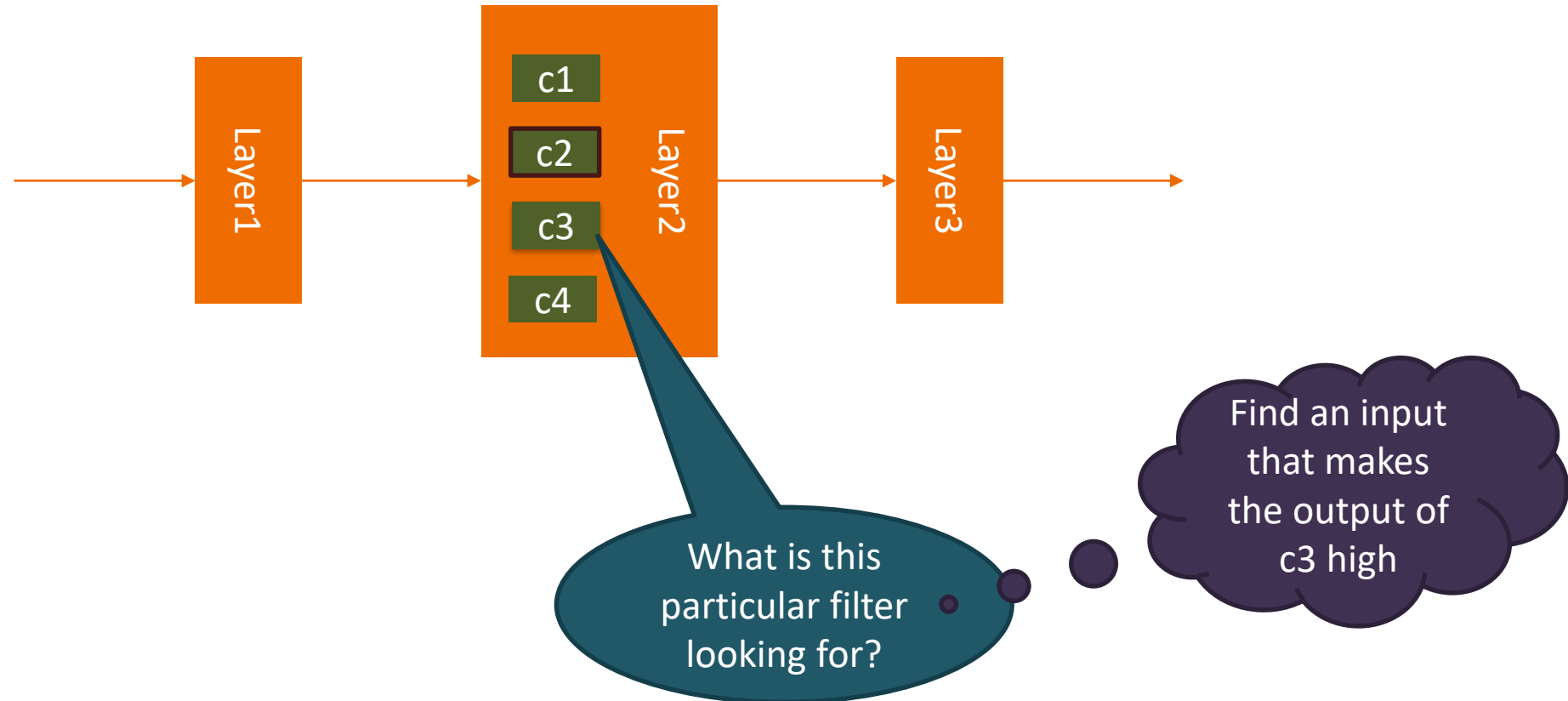


We need to find an image that maximizes the
activation of a filter

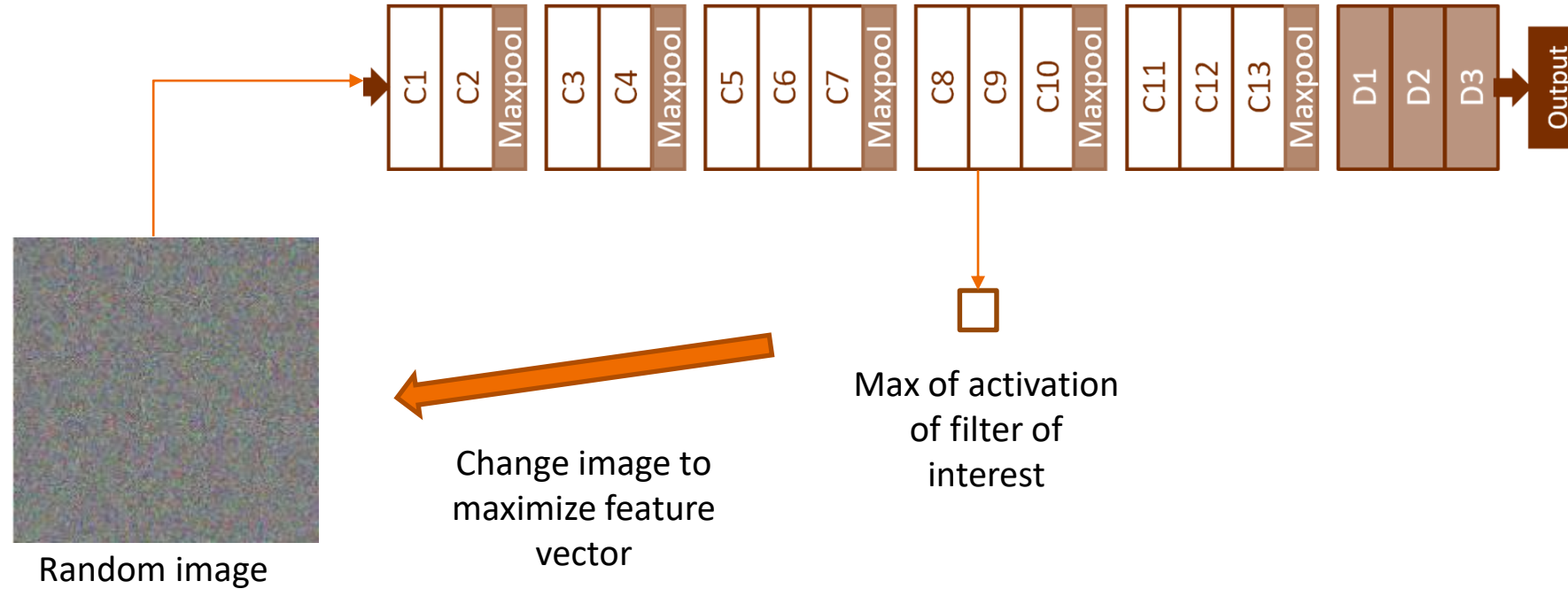
filter_1



Activation maximization

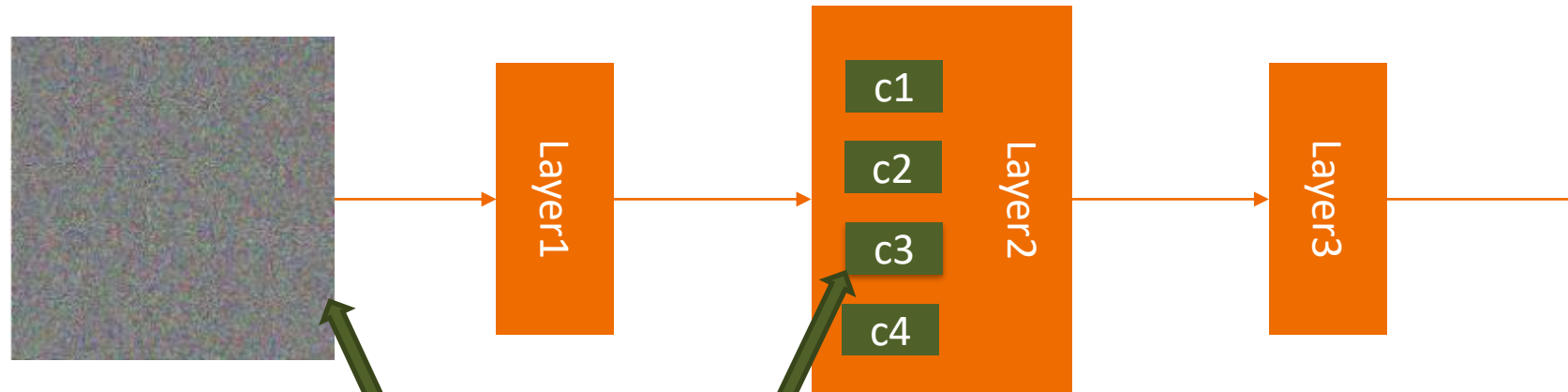


Activation maximization method



- $x^* = \underset{x}{\operatorname{argmax}} h(x) + \lambda R$
- where R is a regularization function

Activation maximization



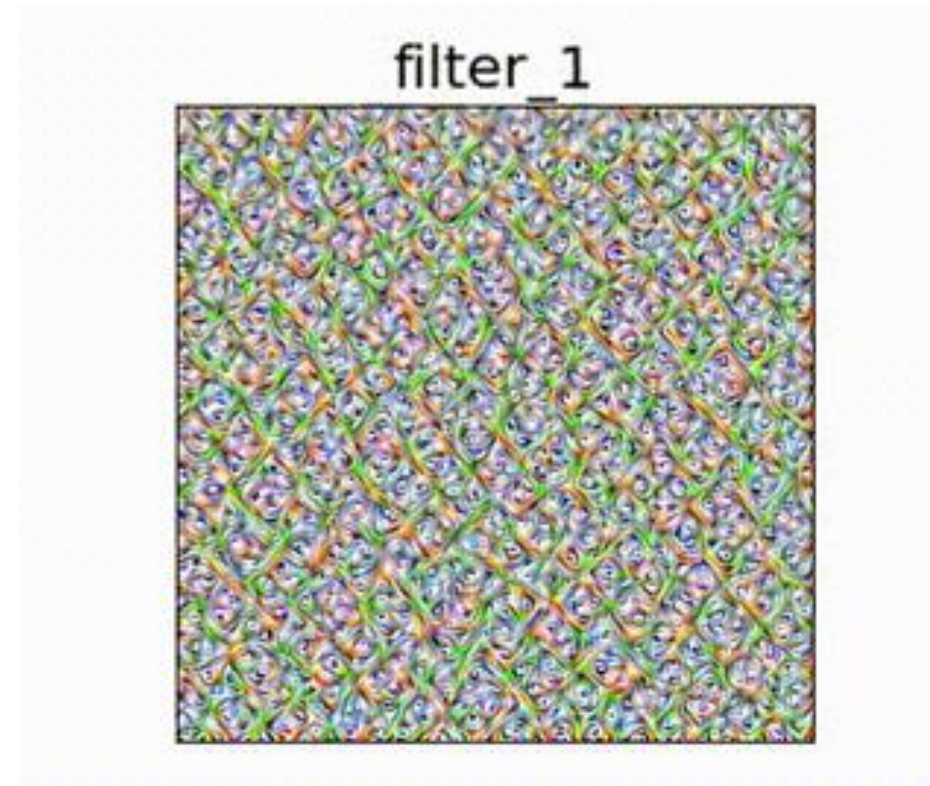
Search through all images
for one that makes the
activation of c3 high

$$\text{Activation} = c3(\text{Layer1}(x))$$

**We can use gradient descent
for this!**

Convolutional Neural Network Filter Visualization

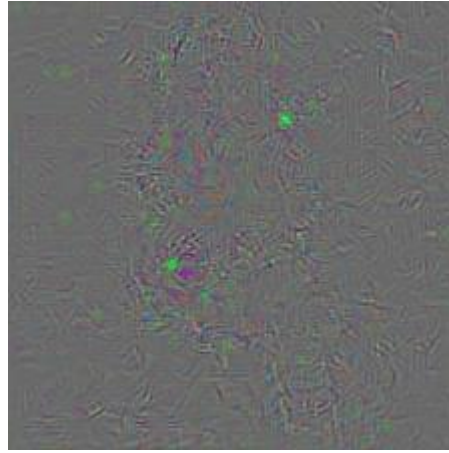
- Optimize the input image with respect to output of the specific convolution operation
- Used a pre-trained **VGG16**
- Visualizations of layers start with basic color and direction filters at lower levels
- Complexity of the filters also increase in the final layers



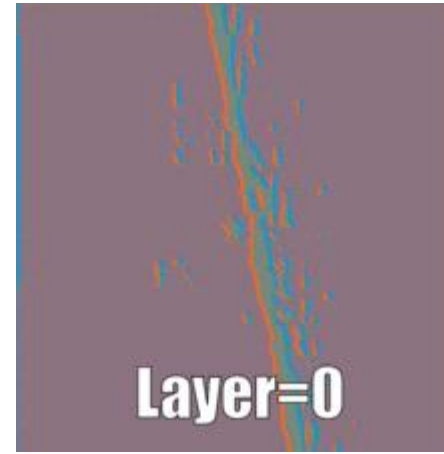
Understanding VGG net w.r.t an input



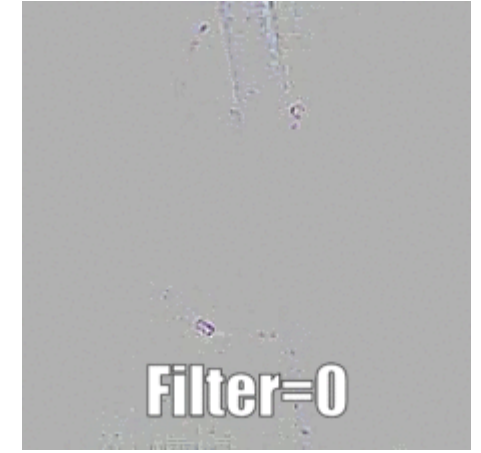
Input Image



Gradients generated
with vanilla back
propagation from Input

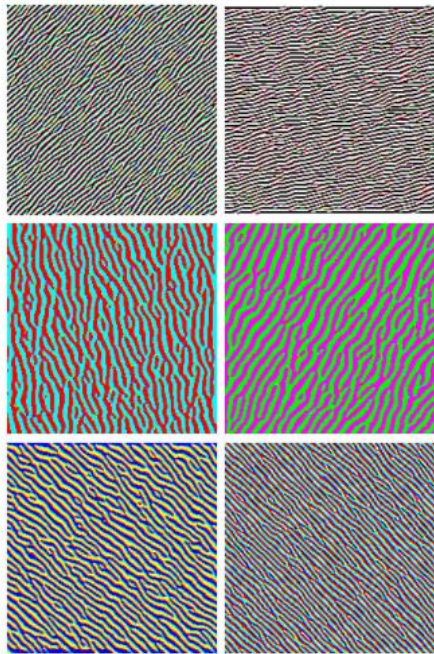


Visualize activations of
Filter 0 in the first 30
layers of VGG16 net for
given input

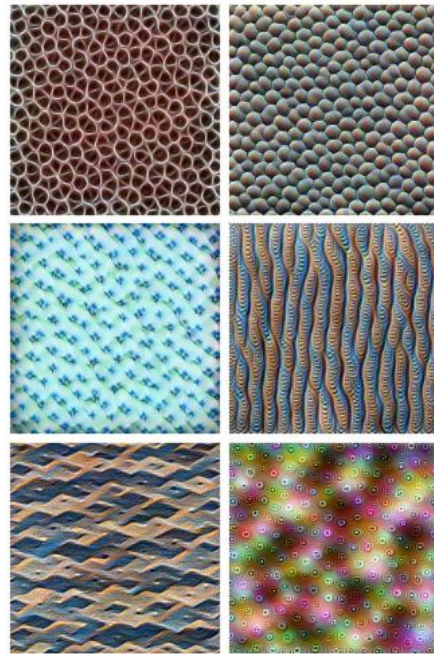


Visualize activations of
the first 30 filters in
layer 29 for given input

Filter hierarchy of object recognition network



Edges (layer conv2d0)



Textures (layer mixed3a)



Patterns (layer mixed4a)

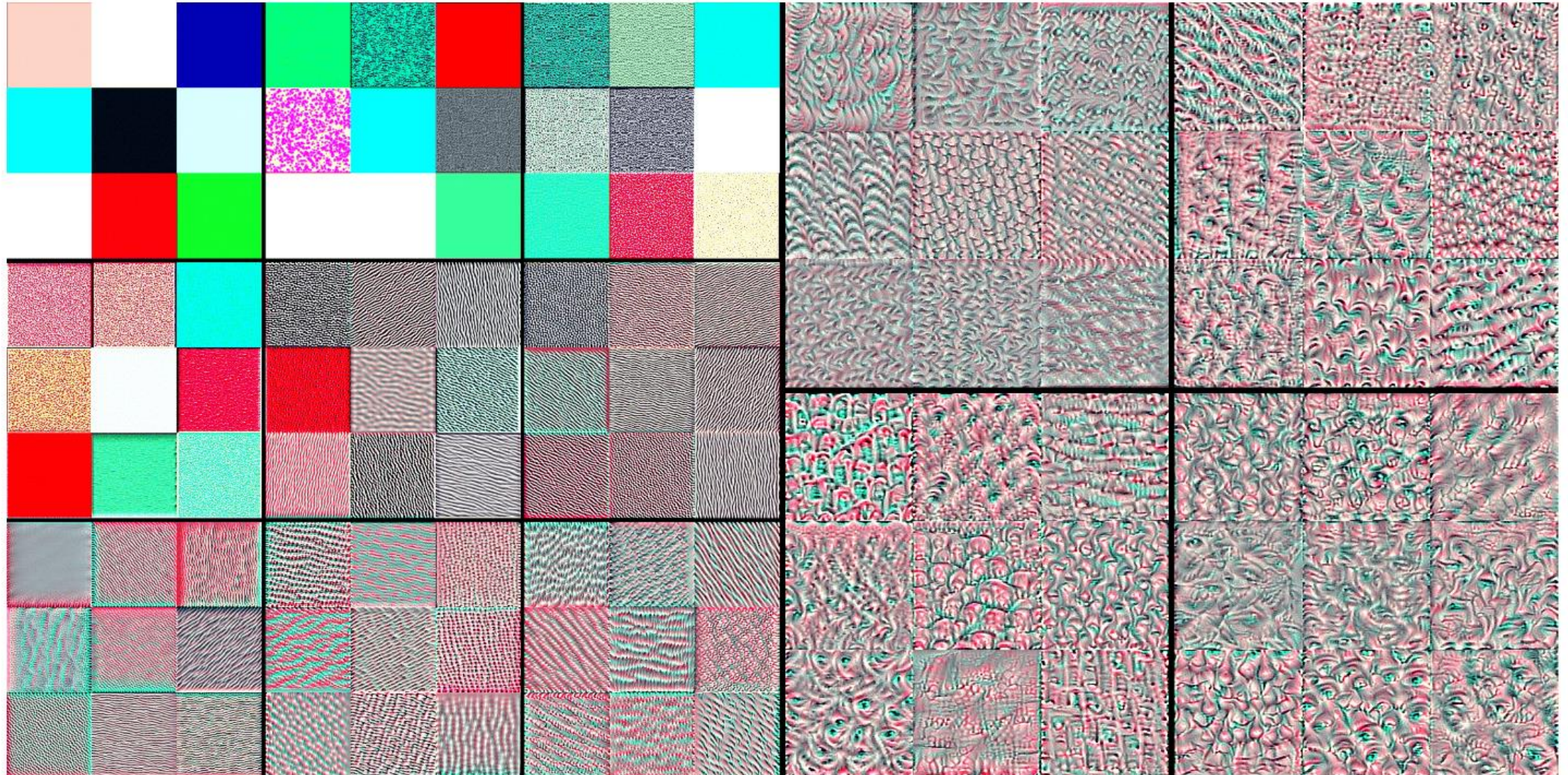


Parts (layers mixed4b & mixed4c)



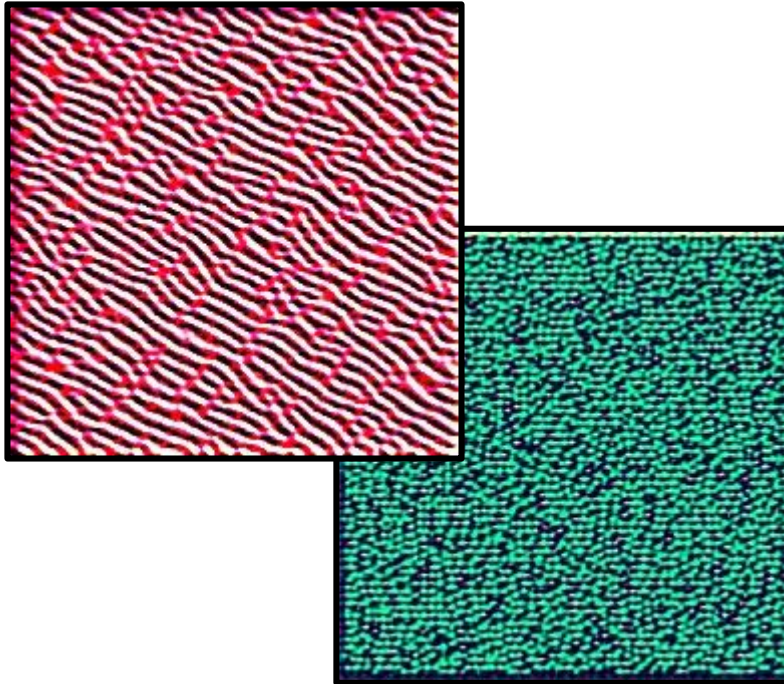
Objects (layers mixed4d & mixed4e)

Filter hierarchy of face networks

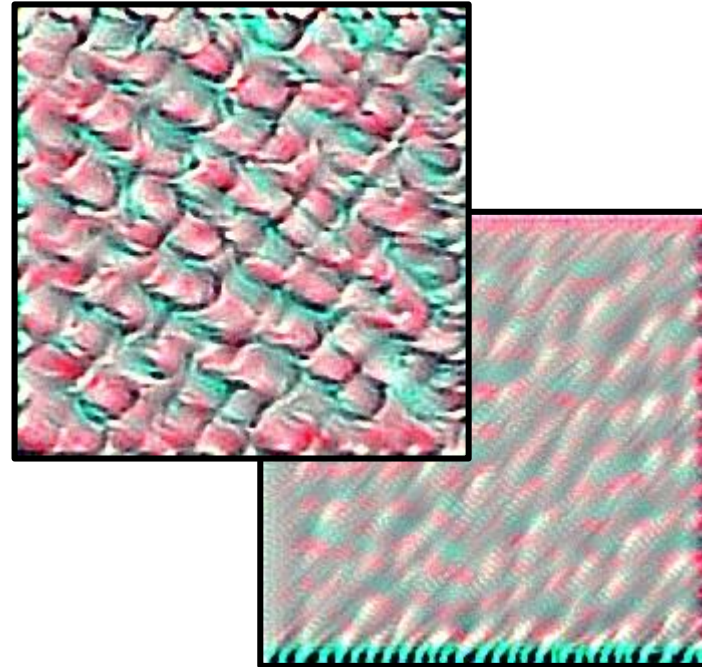


Feature visualization of 9 filters from each conv layer of VGG-Face

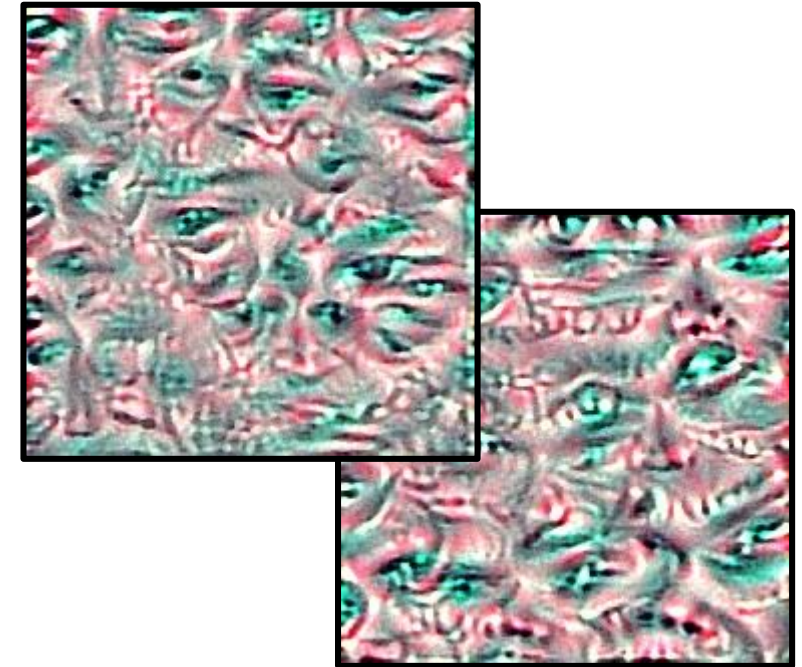
Filter hierarchy for face networks



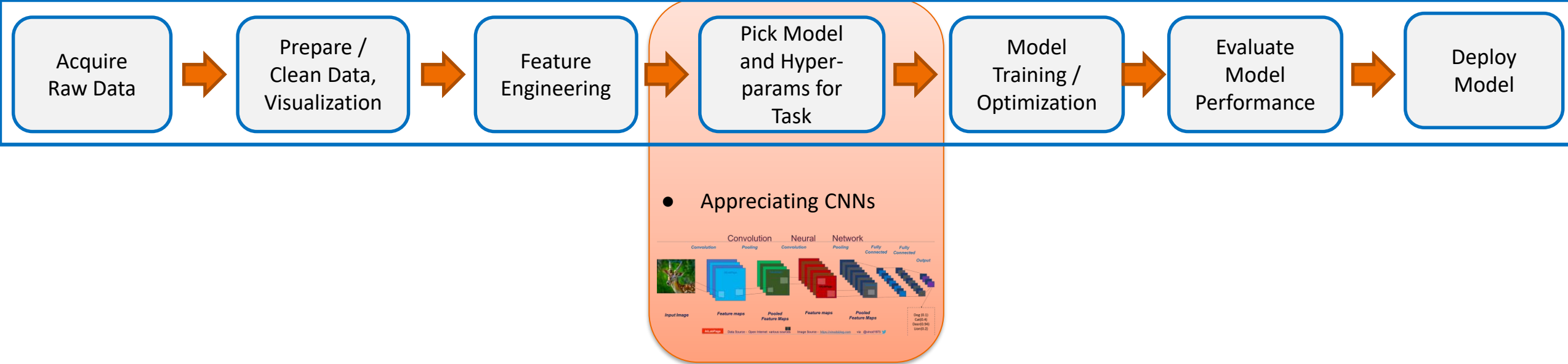
Low-level features



Mid-level features



High-level features



Insight into CNNs

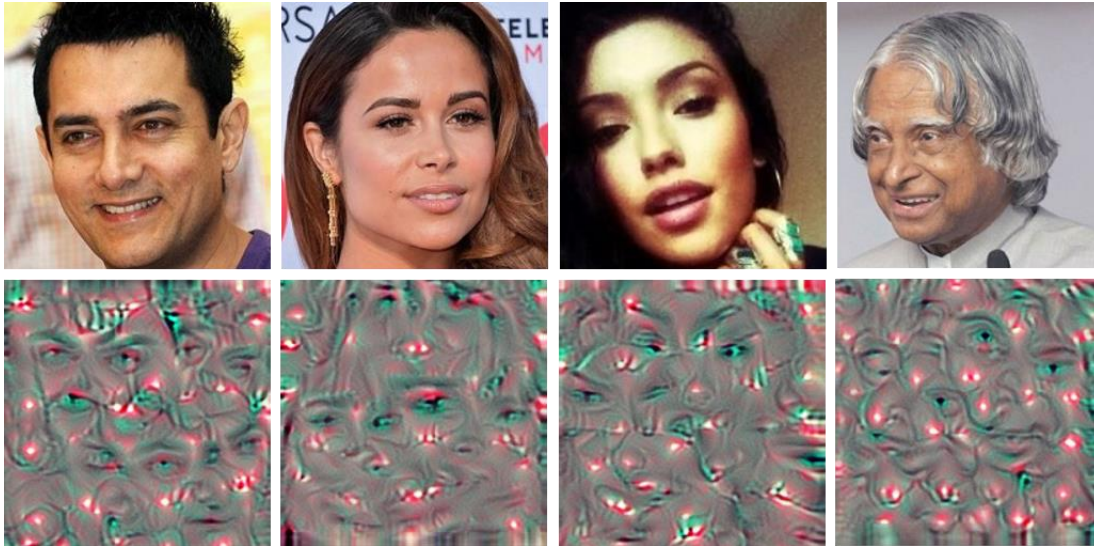
What do they learn?

—

—

Class visualization

- Visualize the output neurons instead of a specific conv filter



Turtle



Tarantula



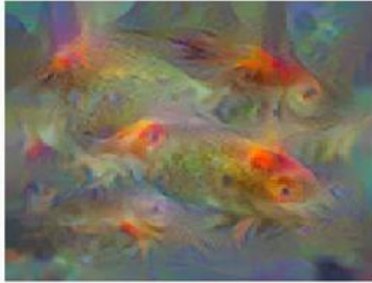
Pelican



Chains

Maximize the **output** of a class neuron

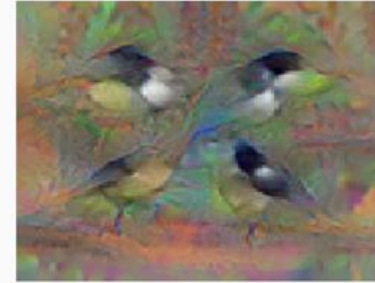
Better Visualizations



(a) Goldfish



(b) Indigo Bunting



(c) Magpie



(d) Kite



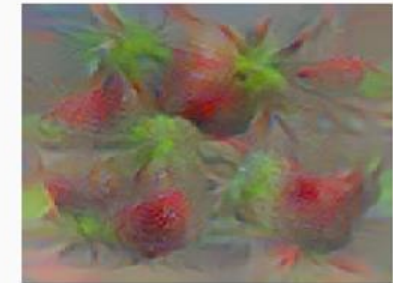
(e) Goose



(f) Flamingo



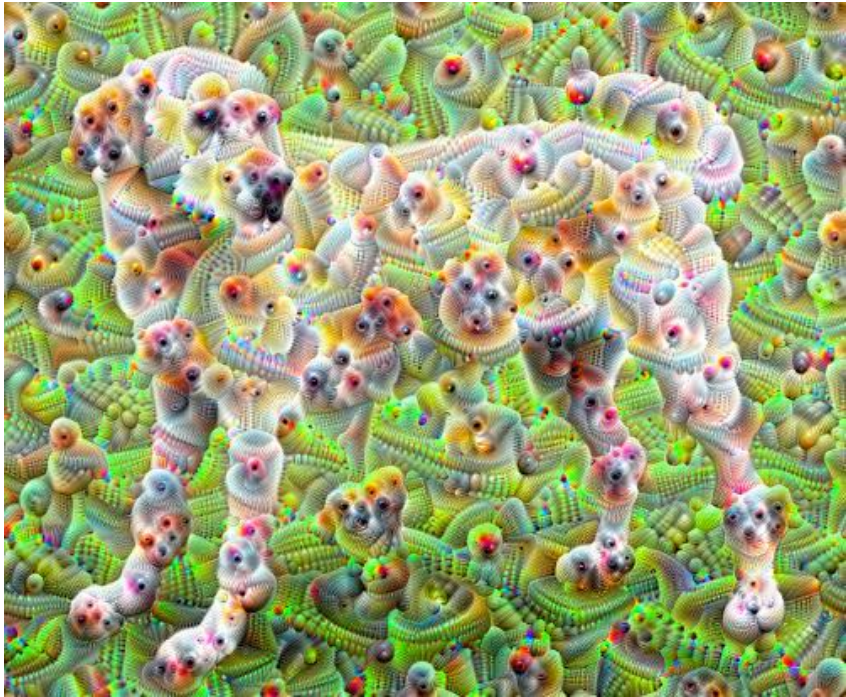
(g) Japanese Spaniel



(h) Strawberry

Deep dream: AI Creativity?

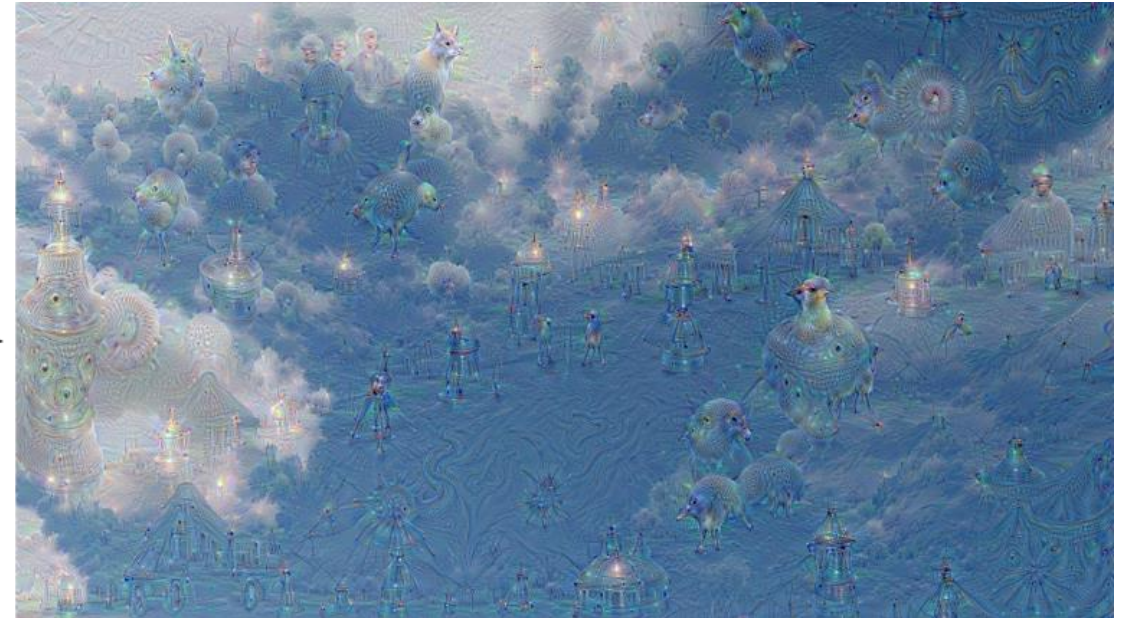
- How does a convnet interpret an image?



<https://www.tensorflow.org/tutorials/generative/deepdream>

Interpreting a cloudy sky

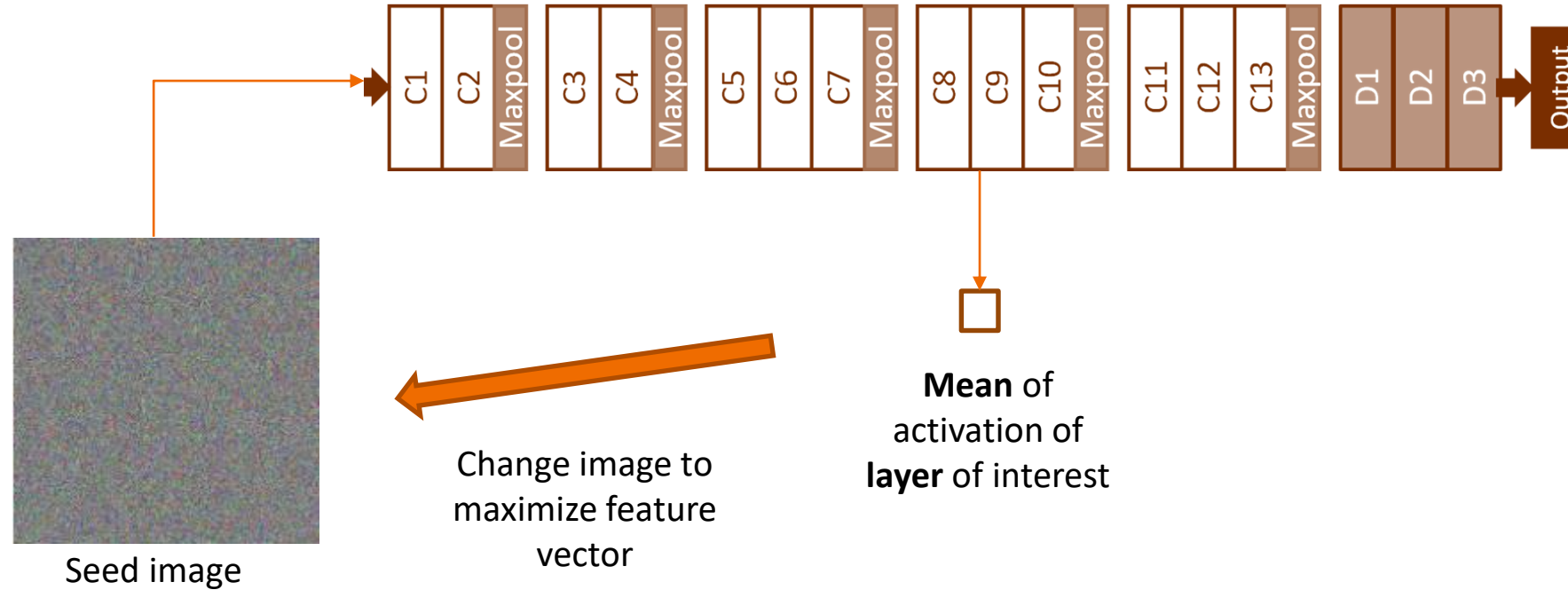
<http://ai.googleblog.com/2015/06/inceptionism-going-deeper-into-neural.html>



Procedure

- Start with a natural image instead of a blank image
- Pick a layer we want to visualize
- Instead of choosing a filter, the loss function is the mean of the entire layer (This makes the model 'choose' what it sees)

Deep Dream method



- $x^* = \underset{x}{\operatorname{argmax}} h(x) + \lambda R$
- where R is a regularization function

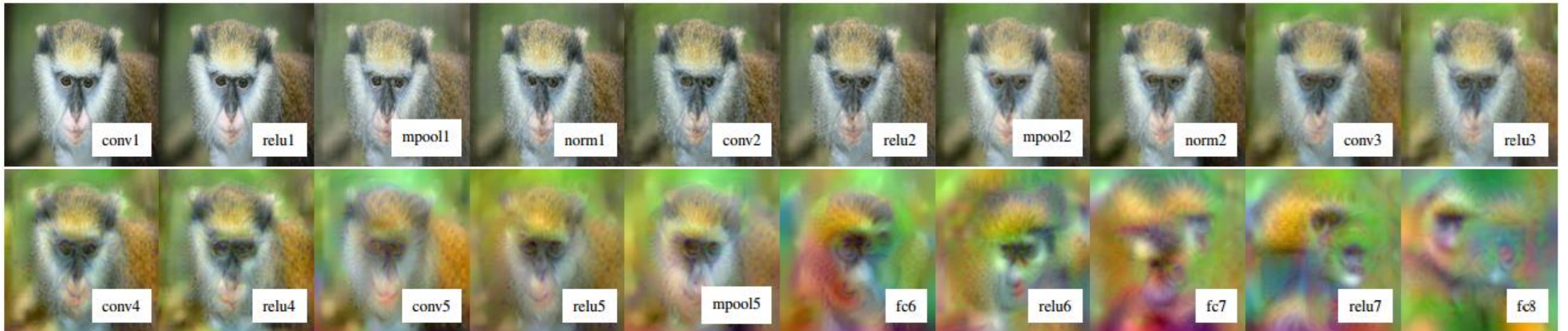
Deep dream: AI Creativity?



<https://deepdreamgenerator.com/>; <https://en.wikipedia.org/wiki/DeepDream>

Feature Inversion

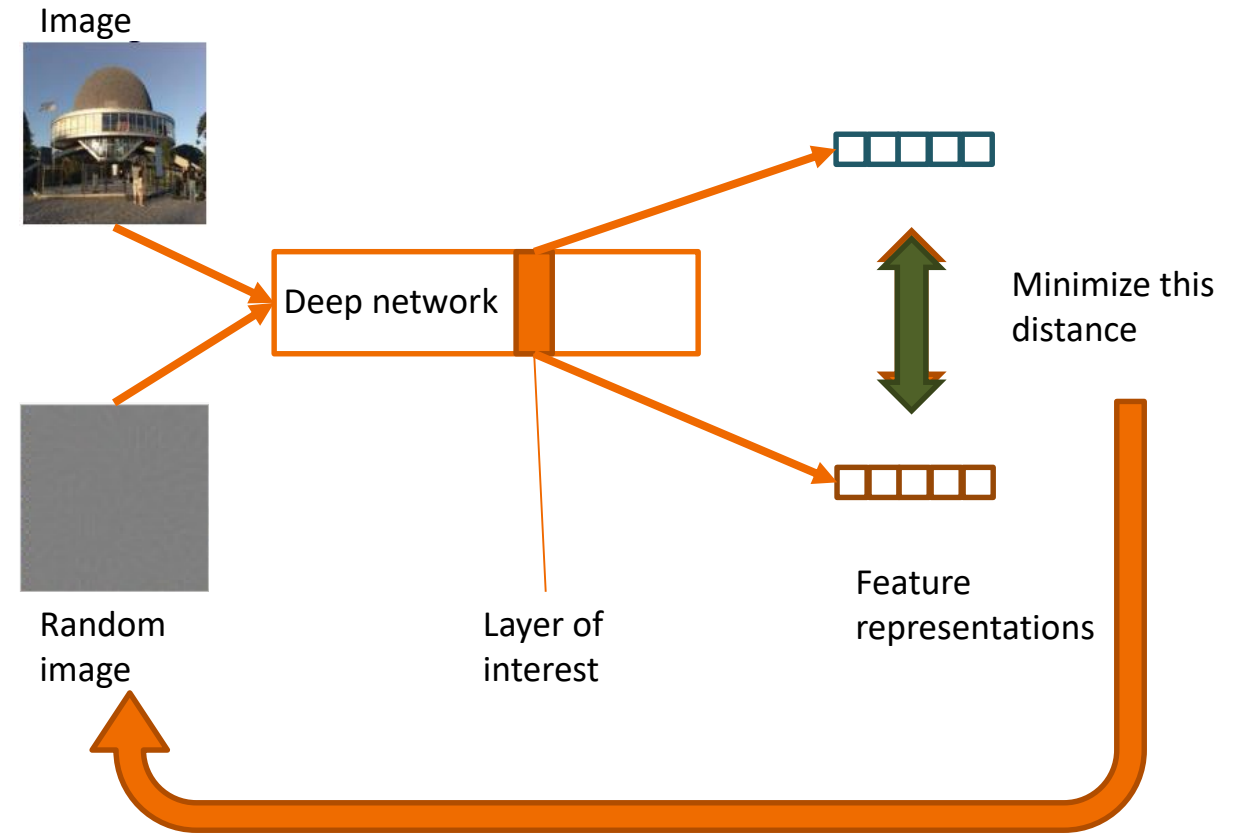
- Invert a feature representation back to image space
- What information is retained or discarded down the layers?



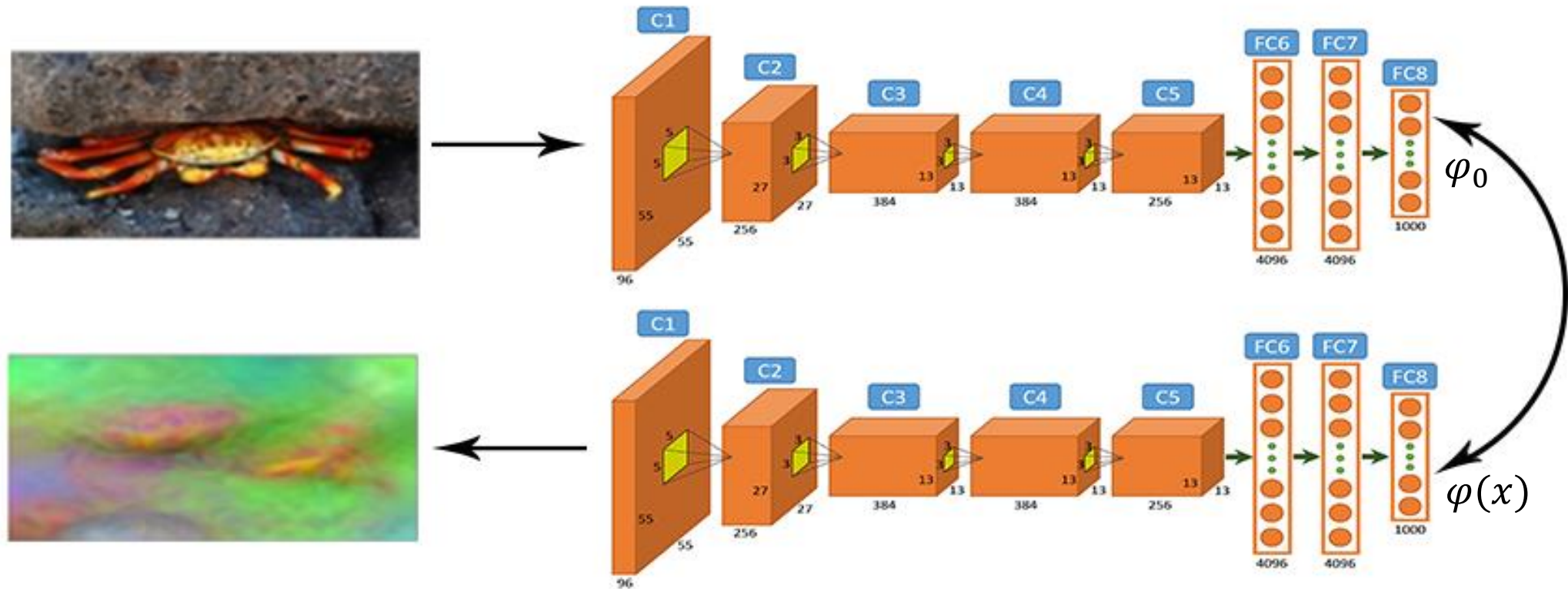
Method

$$\mathbf{x}^* = \underset{\mathbf{x} \in \mathbb{R}^{H \times W \times C}}{\operatorname{argmin}} \ell(\Phi(\mathbf{x}), \Phi_0) + \lambda \mathcal{R}(\mathbf{x})$$

Where $\ell()$ is a distance function between two feature representations and $\mathcal{R}()$ is a regularization function.



Inverting Specific Representation

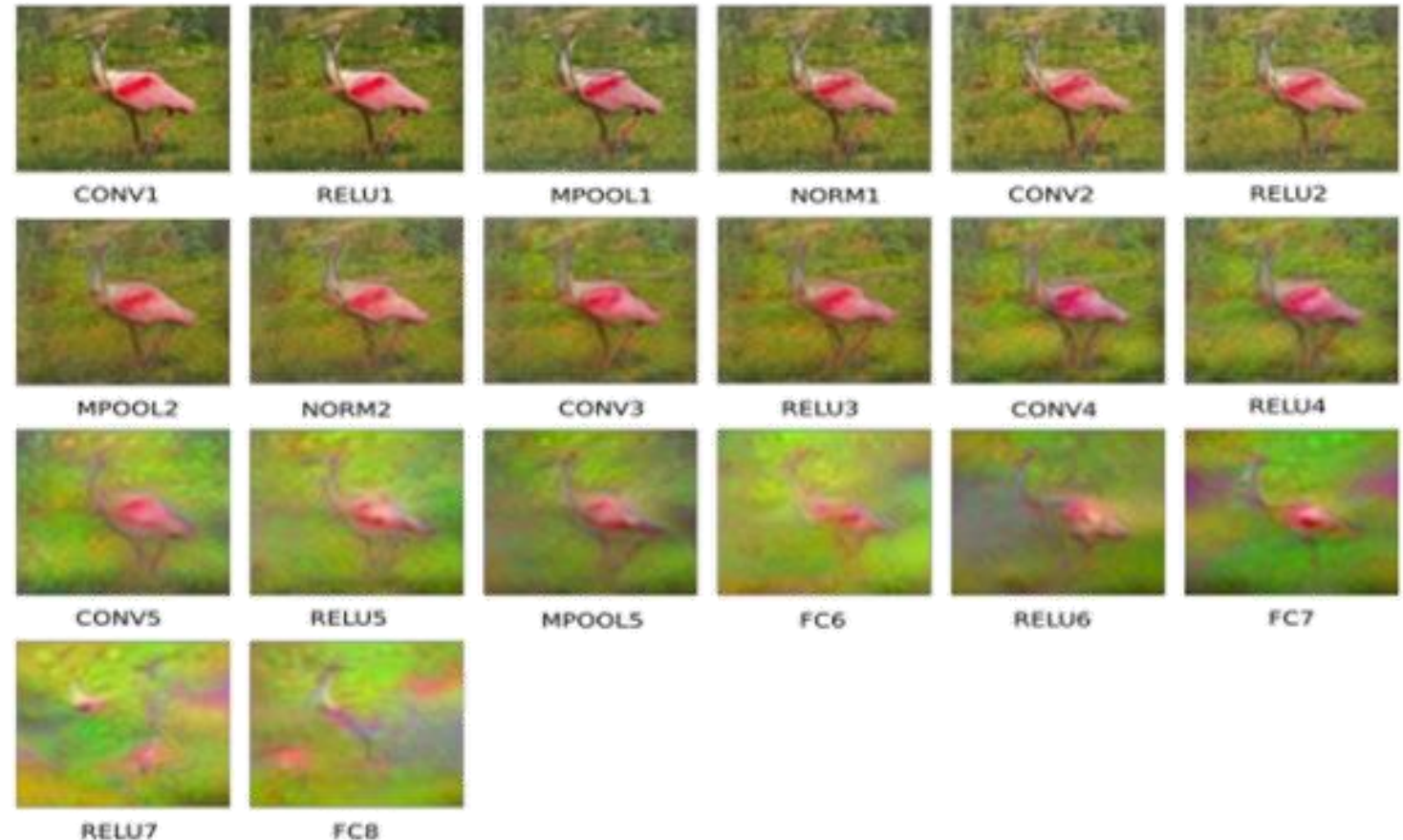


$$x^* = \underset{x \in \mathbb{R}^H \times \mathbb{R}^W \times \mathbb{R}^C}{\operatorname{argmin}} L(\varphi(x), \varphi_0) + \lambda R(x)$$

Aravindh Mahendran and Andrea Vedaldi, Understanding Deep Image Representations by Inverting Them, CVPR'15

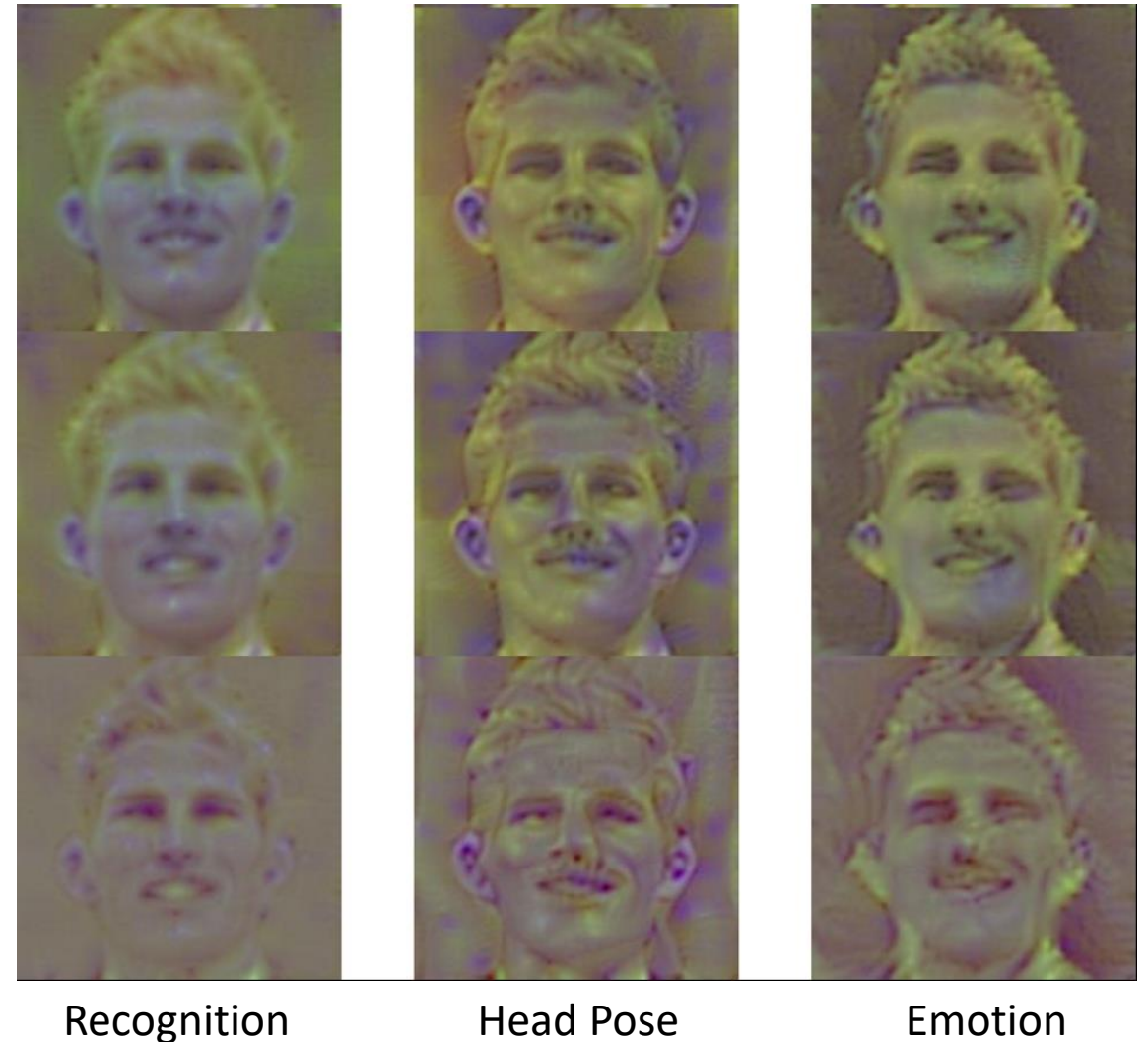
Inverting at different stages

- As we go down the layers, unnecessary information is discarded, and only discriminatory information remains



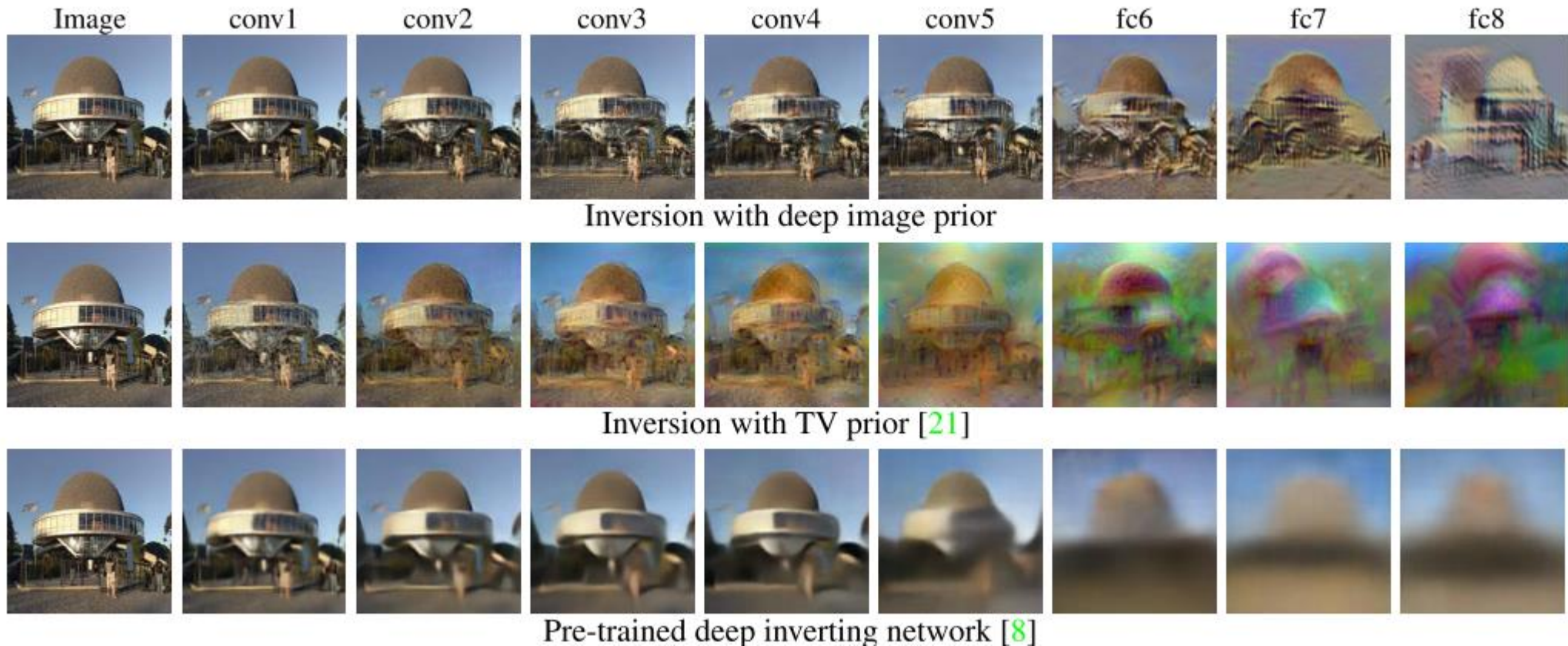
Inverting faces

- Convnets trained for different tasks find different kinds of information useful to keep



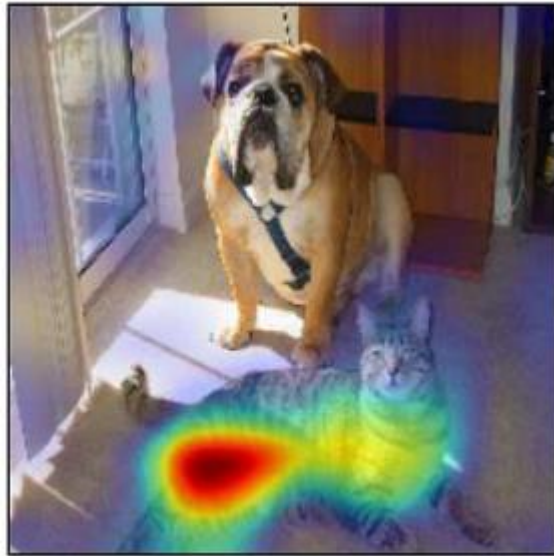
Regularization matters!

- The visualizations are very sensitive to regularization



Saliency Visualization

- Which part of an input image is important?

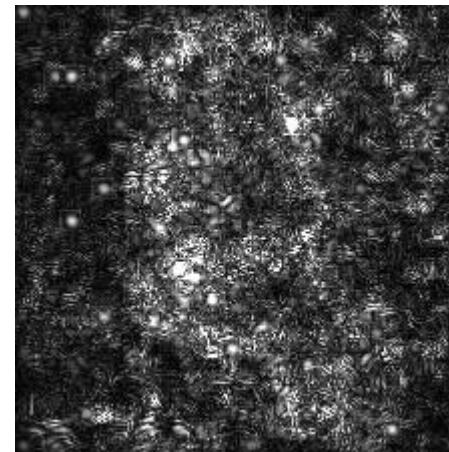
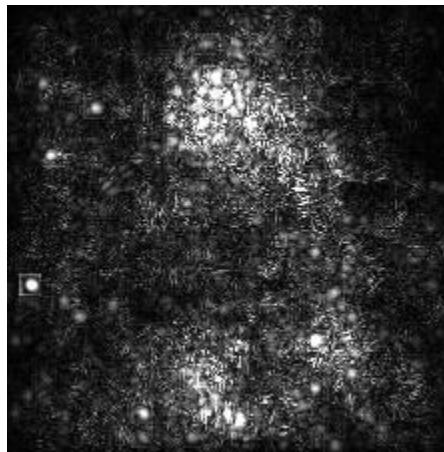
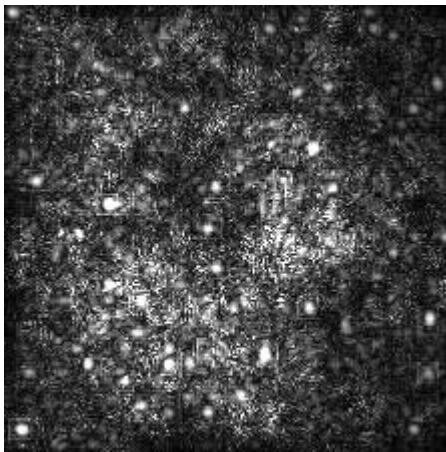


Class is 'cat'

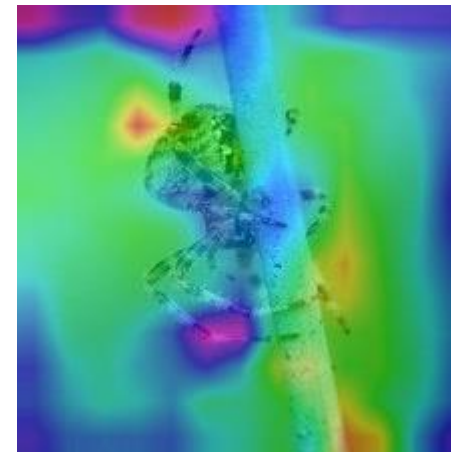
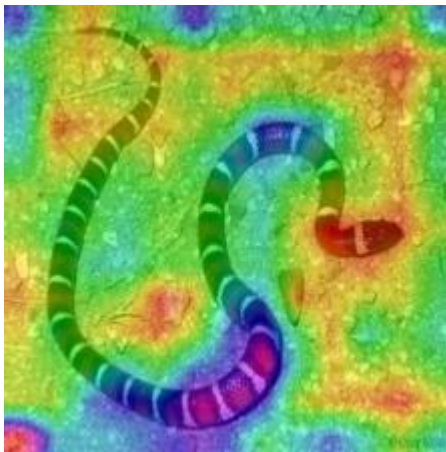


Class is 'dog'

Visualize magnitude of gradients



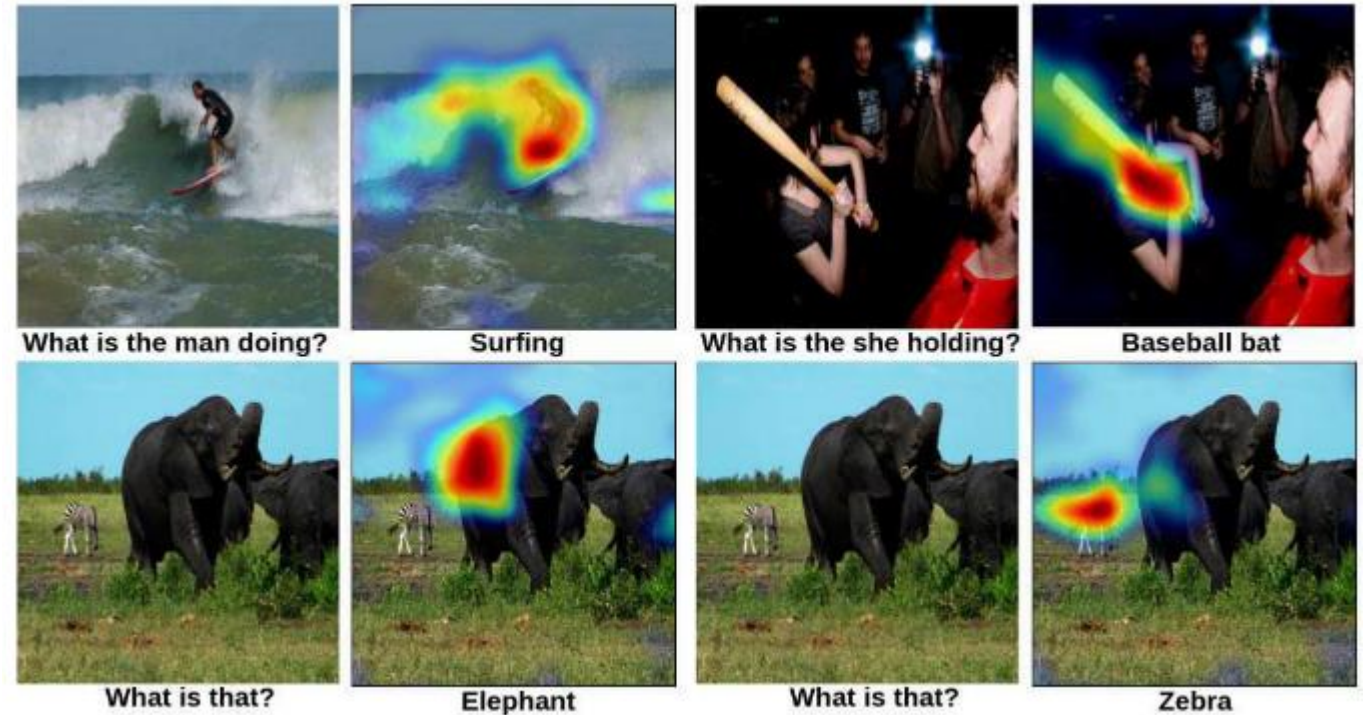
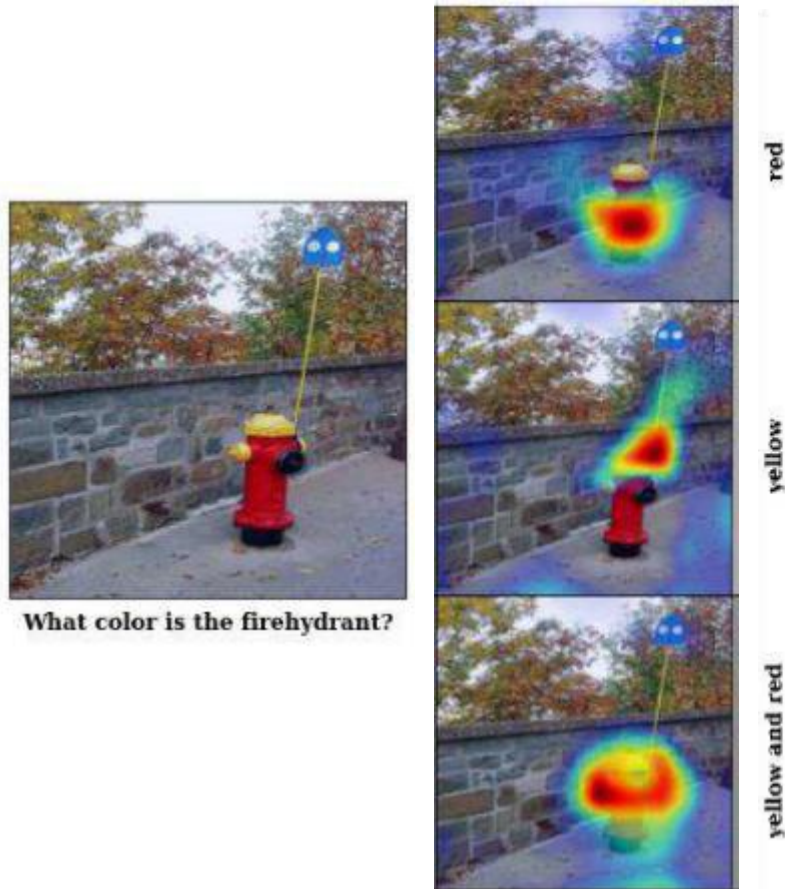
Class activation mapping (Grad-CAM)



Grad-CAM: How

- Two basic Ideas (partly known earlier are combined)
- CAM: Class Activation Maps
 - Limitation to certain type of architectures
- Role of Gradients
 - e.g., Gradient of score of the class wrt to activation from a layer
 - We are interested in feature maps that have positive impact on the class of interest (negative ones correspond to BG or some other class)

GradCAM in Visual Question Answering



Grad-CAM in image captioning



Grad-CAM

Grad-CAM

A group of people flying kites on a beach

A man is sitting at a table with a pizza



A house with a green roof

Sheep grazing in field

A house with a roof

Another: Occlusion map

- Systematically occlude parts of the image and record the drop in classification confidence

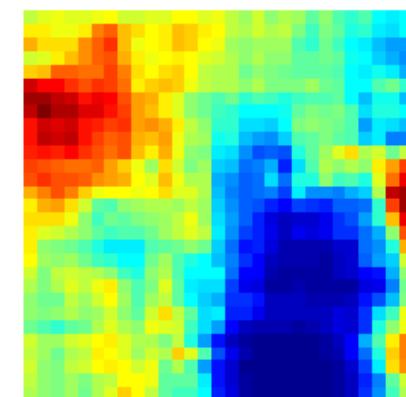
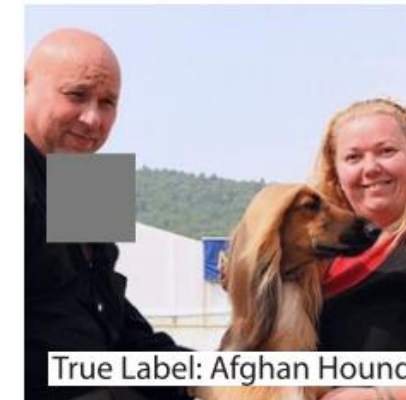
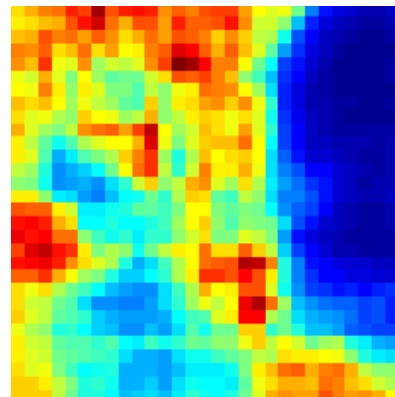
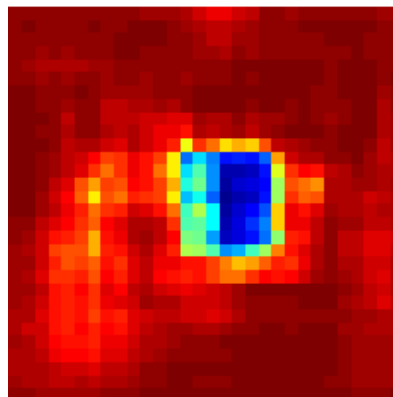
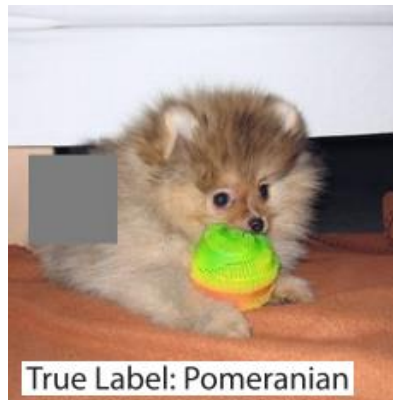
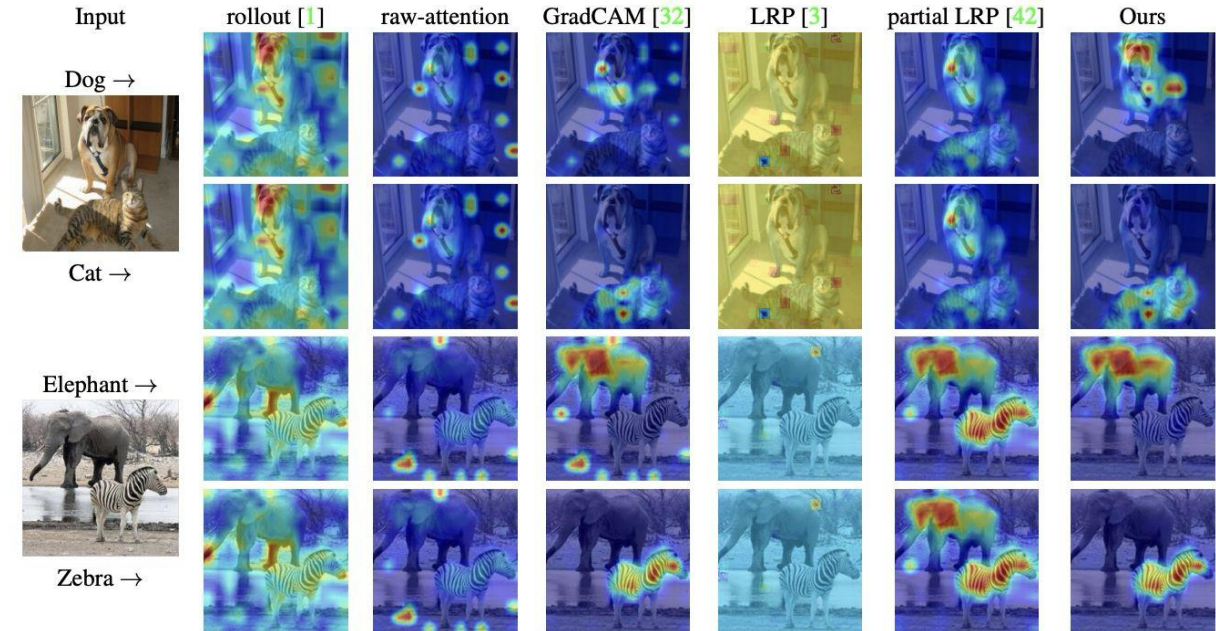
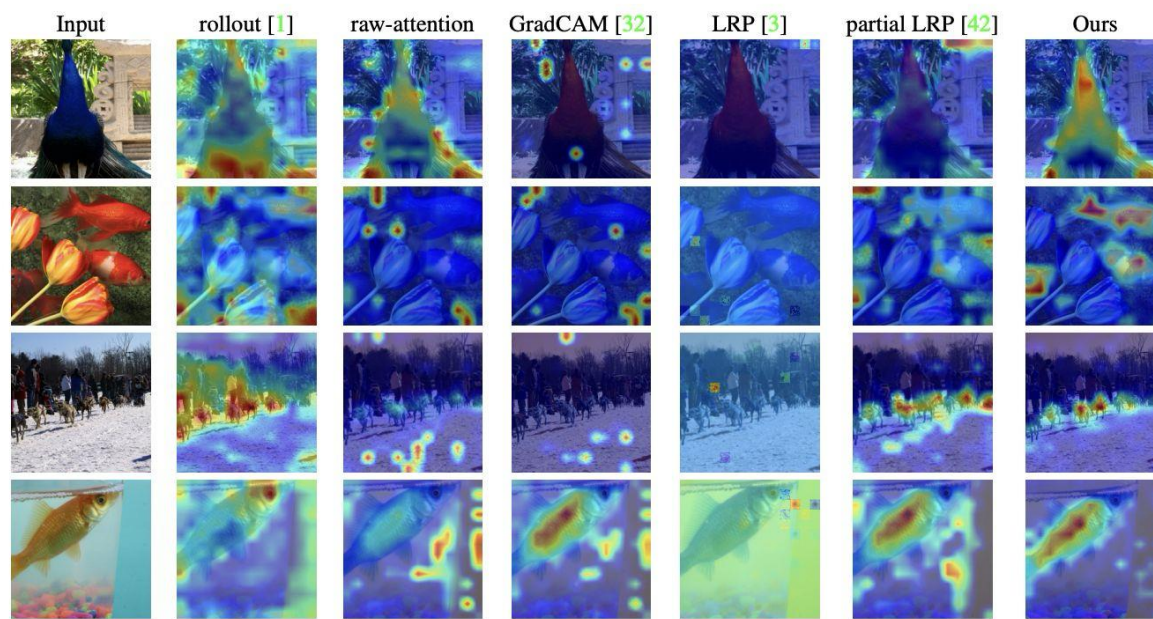


Image source:

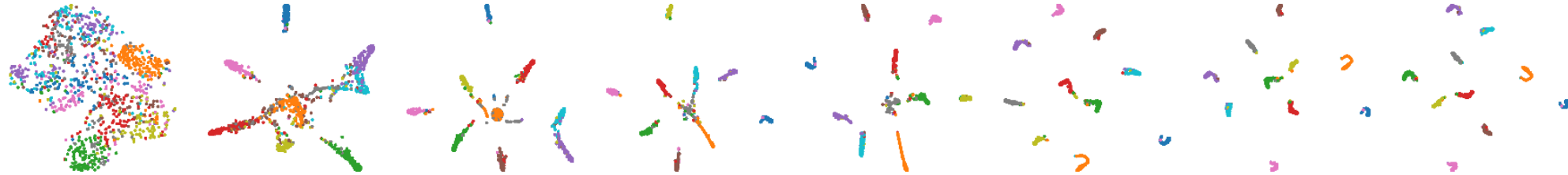
Zeiler, Matthew D., and Rob Fergus. 2014. "Visualizing and Understanding Convolutional Networks. *ECCV 2014*,

Transformer Interpretability



At output: Monitor training dynamics 2D embedding

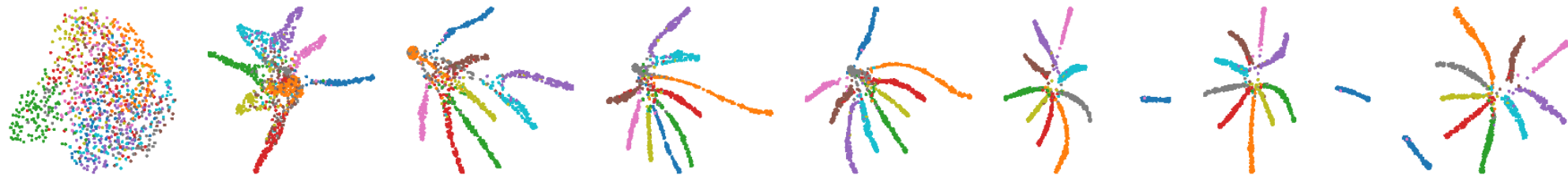
tSne



Dynamic tSne



UMAP



Linear methods



Digit

- 0
- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9

MNIST dataset

Training epochs:

5

10

15

20

30

31

32

Summary

- Visualizing different aspects
- Interpreting for better understanding
- Better explanations
- Better design/refinements

Thanks!!

Questions?

